Exploratory Data Analysis of the Cardiographic Dataset

Introduction

Exploratory Data Analysis (EDA) is a crucial step in the data analysis process, as it helps in understanding the structure, quality, and characteristics of a dataset before applying statistical models or machine learning algorithms. The primary objective of this analysis was to explore the cardiographic dataset, identify data quality issues, understand the distribution of variables, examine relationships between features, and derive meaningful insights that may be useful for future predictive modeling.

The dataset contains information related to fetal cardiotocography measurements, which are commonly used to assess fetal well-being during pregnancy. Various physiological indicators such as fetal heart rate, accelerations, decelerations, uterine contractions, and variability measures are included in the dataset. Through data cleaning, statistical analysis, and visualization techniques, the dataset was examined to gain a better understanding of fetal health patterns and the factors associated with different fetal states.

Dataset Overview

The cardiographic dataset consists of 2,126 observations and 14 variables. Most of the variables represent continuous numerical measurements obtained during fetal monitoring. The target variable in the dataset is NSP, which indicates the fetal state classification. Based on the dataset documentation, NSP contains three valid categories:

- Class 1: Normal
- Class 2: Suspect
- Class 3: Pathologic

The remaining variables describe different physiological characteristics, including baseline fetal heart rate, accelerations, fetal movements, uterine contractions, variability measures, and deceleration-related indicators. Understanding the distribution and relationships among these variables is important for identifying factors that may influence fetal health conditions.

Data Cleaning and Preparation

The first step of the analysis involved examining the dataset for missing values, data type inconsistencies, and potential quality issues.

An initial inspection showed that all variables were stored as numerical values, primarily in floating-point format. Therefore, no major data type conversions were required. However, several columns contained missing values, including LB, AC, DS, DP, MLTV, Width, Tendency, and NSP.

Considering that the number of missing values was relatively small compared to the total dataset size, mean imputation was selected as the most suitable approach. Since all variables were numerical, replacing missing observations with the mean of the corresponding column allowed the dataset size to be preserved while minimizing information loss. This approach was preferred over deleting rows because removing medical records may result in the loss of potentially valuable information. After applying mean imputation, all missing values were successfully handled.

Further investigation of the NSP variable revealed the presence of noisy and invalid category values. According to the dataset description, NSP should contain only three valid classes: 1, 2, and 3. Some observations contained values such as -1 and 5, which do not correspond to any valid fetal state category. Because NSP represents a class label rather than a continuous variable, replacing invalid values through imputation would not be appropriate. Instead, these invalid records were removed from the analysis. The NSP column was then converted into a categorical variable to better reflect its role as a classification target.

Outlier detection was performed using boxplots and the Interquartile Range (IQR) method. The analysis initially identified a large number of potential outliers across several variables. However, a closer examination revealed that many features such as AC, FM, UC, DL, DS, and DP contain numerous values close to zero. In addition, the statistical summary showed that the mean, minimum, and maximum values of several variables are clustered within a relatively narrow range around zero.

Since the dataset belongs to the medical domain, these extreme values may represent genuine physiological measurements rather than data entry errors. Removing them could lead to the loss of important clinical information and potentially bias future analyses. Therefore, the identified outliers were retained for subsequent analysis.

Statistical Summary

Descriptive statistics were calculated for all numerical variables to understand their central tendency and variability.

The baseline fetal heart rate (LB) had an average value of approximately 133.34 beats per minute with a median value of 133. This indicates that the distribution of LB is relatively balanced and falls within a normal physiological range. The standard deviation of 11.27 suggests moderate variability among observations.

Several variables, including AC, FM, DL, DS, and DP, exhibited very small average values and median values close to zero. For example, the median values of FM, DS, and DP were all equal to zero. This indicates that these events were absent in more than half of the observations and occur relatively infrequently within the dataset.

The variability measures showed notable differences across observations. ASTV had a mean value of approximately 47 with a standard deviation of 18.81, while ALTV had a mean value of 10.29 and a standard deviation of 21.21. The relatively large standard deviations indicate considerable variation in these measurements among different fetal monitoring sessions.

A comparison of mean and median values suggests that some variables, such as LB, UC, MSTV, and Width, have relatively balanced distributions. However, several other variables demonstrate varying degrees of skewness. This observation is supported by the visual analysis performed using histograms.

The statistical summary also highlighted the presence of wide value ranges in variables such as Width, ALTV, MLTV, and ASTV, indicating substantial differences between observations. The standard deviation and interquartile range measurements further confirmed that variability levels differ significantly across the dataset.

Overall, the descriptive analysis revealed that the dataset contains a mixture of moderately distributed variables and highly skewed variables with many zero values, which is consistent with the nature of physiological monitoring data.

Data Visualization and Findings

Several visualization techniques were used to explore the distribution of variables and identify underlying patterns within the dataset.

Histograms were generated for all numerical variables. Most variables exhibited right-skewed distributions, with a large concentration of observations near zero. This pattern was particularly noticeable for variables such as AC, FM, DL, DS, and DP. The presence of numerous zero values explains why these variables were frequently identified as outliers during IQR-based analysis.

In contrast, the LB variable displayed a more symmetric distribution around its mean value, indicating a relatively stable range of baseline fetal heart rates across observations.

The distribution of the target variable NSP was visualized using a bar chart. The analysis showed the following class frequencies:

- Normal (NSP = 1): 1,651 observations
- Suspect (NSP = 2): 293 observations
- Pathologic (NSP = 3): 172 observations

The results clearly indicate that the dataset is imbalanced, with the Normal class accounting for the majority of observations. The Suspect and Pathologic classes are considerably underrepresented in comparison.

At this stage, no balancing techniques were applied because the objective of the assignment was exploratory analysis rather than predictive modeling. Furthermore, class balancing should not be performed automatically without evaluating its impact on model performance. Future machine learning models may benefit from techniques such as class weighting, oversampling, undersampling, or synthetic sample generation if the imbalance negatively affects predictive performance.

A correlation heatmap was also generated to examine relationships among variables. The heatmap provided a useful overview of how strongly variables are related to one another and helped identify potential predictors of fetal health status.

Pair plots were explored as an advanced visualization technique. However, because the dataset contains numerous numerical variables, the resulting plots became visually crowded and difficult to interpret. Since the correlation heatmap already provided a clear overview of feature relationships, additional pairwise analysis was not pursued further.

Correlation Analysis and Pattern Recognition

Correlation analysis was performed to identify variables that may influence fetal health status represented by the NSP target variable.

The strongest positive correlation with NSP was observed for ASTV (r = 0.423). This was followed by ALTV (r = 0.366) and DP (r = 0.294). These findings suggest that abnormal short-term variability, abnormal long-term variability, and prolonged decelerations may be important indicators of fetal health conditions. Higher values of these variables appear to be associated with increasing severity in fetal state classification.

Among all variables, AC demonstrated the strongest negative correlation with NSP (r = -0.317). This suggests that higher acceleration activity is generally associated with healthier fetal conditions and lower-risk fetal states.

Several additional variables showed weaker relationships with NSP. For example, UC exhibited a negative correlation of -0.185, while LB showed a positive correlation of 0.132. Although these relationships are weaker, they may still contribute useful information when combined with other features during predictive modeling.

The correlation matrix also revealed interesting relationships among predictor variables. Moderate positive correlations were observed between MSTV and Width (r = 0.435), DL and MSTV (r = 0.410), DL and Width (r = 0.410), and ASTV and ALTV (r = 0.357). These relationships indicate that certain physiological measurements tend to vary together.

Despite the presence of these moderate associations, most feature pairs exhibited relatively weak correlations. This suggests that multicollinearity is generally low within the dataset, meaning that most variables provide unique information rather than duplicating the information contained in other features.

The assignment also required identifying trends over time whenever temporal data was available. However, the dataset does not contain any time-related variables or timestamps. Therefore, trend analysis and time-series pattern identification were not possible.

Key Insights

Several important findings emerged from the exploratory analysis:

The dataset contains only a small proportion of missing values, making mean imputation an effective method for preserving the dataset while handling incomplete observations.

Many variables contain a substantial number of zero values, which contributes to the skewed distributions observed in the histograms and explains the large number of observations flagged as outliers during IQR analysis.

The target variable NSP is significantly imbalanced, with the majority of observations belonging to the Normal fetal state category. This imbalance should be considered during future machine learning model development.

ASTV, ALTV, and DP showed the strongest positive relationships with fetal state classification, suggesting that they may serve as important predictive features.

AC demonstrated a moderate negative relationship with NSP, indicating that higher acceleration activity is generally associated with healthier fetal conditions.

Most variables displayed weak correlations with one another, suggesting low multicollinearity and indicating that many features contribute distinct information.

Recommendations for Future Analysis

The findings from this exploratory analysis provide a strong foundation for future predictive modeling and healthcare-related investigations.

Future work could involve feature selection techniques to identify the most influential predictors of fetal health status. Variables such as ASTV, ALTV, DP, and AC should receive particular attention because of their stronger relationships with the target variable.

Before developing classification models, the impact of class imbalance should be carefully evaluated. If necessary, techniques such as class weighting, oversampling, undersampling, or synthetic sample generation methods may be applied to improve the model's ability to correctly identify minority classes.

Additional model evaluation metrics beyond overall accuracy, such as precision, recall, F1-score, and confusion matrices, should also be considered because they provide a more reliable assessment when working with imbalanced datasets.

Conclusion

This exploratory data analysis provided a comprehensive understanding of the cardiographic dataset and its underlying characteristics. The data cleaning process successfully addressed missing values, removed invalid target labels, and prepared the dataset for analysis while preserving valuable medical information.

Descriptive statistics and visualizations revealed that many variables exhibit skewed distributions and contain large numbers of zero values. These characteristics explain the high number of observations identified as outliers and justify the decision to retain them for analysis. The target variable NSP was found to be noticeably imbalanced, with Normal fetal states dominating the dataset.

Correlation analysis identified ASTV, ALTV, and DP as the variables most strongly associated with fetal state classification, while AC demonstrated an inverse relationship with NSP. Most other variables exhibited relatively weak correlations, indicating low multicollinearity and suggesting that the features provide complementary information.

Overall, the analysis successfully achieved its objective of exploring the dataset, identifying important patterns, and uncovering relationships among variables. The insights obtained from this study can support future machine learning applications, feature selection efforts, and predictive modeling aimed at assessing fetal health conditions more effectively.